

ADITYA VADALI

B.Tech in CSE & MS in CSE (By Research), IIIT Hyderabad | CGPA: 8.04

github.com/CoderKai9001 | moulichandra4321@gmail.com

EDUCATION

International Institute of Information Technology, Hyderabad (IIITH)	2023 – Present
B.Tech & MS (By Research) in Computer Science and Engineering	CGPA: 8.04

RESEARCH EXPERIENCE

Undergraduate Researcher	2024 – Present
<i>Robotics Research Center, IIIT Hyderabad</i>	<i>Advisor: Prof. K. Madhava Krishna</i>
· Robotic perception, 3D vision, and planning & navigation. · Vision-based navigation and 3D reconstruction pipelines, deployed on physical robot platforms.	

PUBLICATIONS

PixelLoop: Shortcut Topological Navigation with Pixel-Level Loops [arXiv]	Accepted, IROS 2026
Plann3r: Predicting Planning Costs Grounded in 3D	Accepted, and will be presenting in CoRL 2026

PROJECTS

Splatt3r-SfM GitHub	2025
<i>3D Vision</i>	
· Coupled MAST3R-SfM N-view pose estimation with Splatt3r's 2-view Gaussian head, fusing pairwise splats into one aligned scene. · Feed-forward novel view synthesis from unposed image collections, with no per-scene optimization.	
NoMaD Deployment on P3DX GitHub	2025
<i>Robot Navigation</i>	
· Deployed an image-goal navigation model on a Pioneer P3DX over ROS for closed-loop control in real environments. · Optimized the inference path to cut per-step latency and sustain real-time onboard operation.	
Ray Tracer in C++ GitHub	2024
<i>Computer Graphics</i>	
· Analytic ray-primitive intersection for spheres, cylinders, cuboids, ellipsoids and cones. · Full pipeline: camera rays, lighting, shading and shadows, rendered to an OpenGL display.	

HACKATHONS & ACHIEVEMENTS

- **1st – Megathon 2025, IIITH:** [SnapGen](#) – text+image Stable Diffusion editor running fully on-device on the QIDK NPU.
- **2nd – Megathon 2024, IIITH:** [Yoga app](#) – on-device keypoint detection with real-time pose-correction feedback.
- **3rd – Lloyds Banking Group IP&I, Dec 2025:** [agentic system](#) detecting and repairing data inconsistencies on Google Cloud.
- **Most Impactful Project – HackIIIT, Feb 2026:** mess crowd-prediction system live on the MyIIIT app; used by 1000+ IIIT students today.

SKILLS

Languages	Python, C, C++, JavaScript, TypeScript, SQL
Frameworks	PyTorch, TensorFlow, ScikitLearn, React, Next.js, Node.js, Flask, FastAPI
Tools	Docker, ROS, Git, OpenGL, SSH, NumPy, Pandas
Domains	Deep Learning, Computer Vision, 3D Vision, Robotic Perception, Planning & Navigation